

French Financial Services Sector — Multi-Cloud Transformation Programs 2023-2026

BEFORE: Legacy State (Pre-2023)

- AFTER: Target State (End 2026)**

- AWS (primary): 3 regions (eu-west-1, eu-west-3, eu-central-1)
- Azure (secondary): compliance workloads, M365 integration
- 78% containerized (EKS), 15% serverless (Lambda), 7% EC2
- Deployment frequency: 12x/day (target: 50x/day by 2026)
- Infrastructure cost: EUR 540M/year (target, -39%)
- Cloud-native team: 1,800 engineers (retrained + new hires)
- MTTR: 18 minutes (target)
- Security: zero-trust, AWS GuardDuty + Prisma Cloud
- Compliance: automated via AWS Config + custom rules
- Database: Aurora PostgreSQL, DynamoDB, Redis ElastiCache
- Technical debt score: 3.1/10 (target)
- Change failure rate: 5% (target)

"La migration cloud n'est pas un projet IT, c'est une transformation business. Nous avons redéfini notre modèle opérationnel autour du cloud, avec l'objectif de déployer 80% de nos workloads critiques sur AWS d'ici fin 2026. Cela représente un investissement de plus de 700 millions d'euros sur 4 ans, mais le ROI est évident : réduction des coûts d'infrastructure de 39%, time-to-market divisé par 10." — Philippe VALLEE, Directeur des Systèmes d'Information, Societe Generale (interview Le Monde Informatique, Sept. 2025)

- AWS Solutions Architects (x45): Design multi-account landing zone, AWS Organizations, Control Tower. Salary: 65-95k EUR.
- Platform Engineers / SRE (x80): Terraform, Crossplane, ArgoCD, EKS management. Salary: 55-80k EUR.
- Cloud Security Engineers (x35): AWS SecurityHub, GuardDuty, IAM policies, CSPM. Salary: 60-85k EUR.
- Data Engineers (x60): AWS Glue, EMR, Redshift, Lake Formation, Kafka/MSK. Salary: 55-85k EUR.
- DevOps / CI-CD Specialists (x50): Jenkins!GitLab CI migration, artifact management, quality gates. Salary: 50-70k EUR.
- Cloud FinOps Analysts (x15): AWS Cost Explorer, Kubecost, Reserved Instances optimization. Salary: 50-65k EUR.
- Remaining ~115 positions: project managers, Scrum masters, change management, training.

Technology stack detail (Architecture Decision Records — ADR-2024-037 through ADR-2024-089):
 Compute: Amazon EKS (Kubernetes 1.28+), AWS Lambda, AWS Fargate
 Networking: AWS Transit Gateway, PrivateLink, Direct Connect (2x 100Gbps to Fontenay DC)
 Storage: S3 (IA/Glacier for compliance archives), EFS, FSx for Lustre (quant models)
 Database: Aurora PostgreSQL 15, DynamoDB, ElastiCache Redis 7, Amazon Timestream (market data)
 Messaging: Amazon MSK (Kafka), SQS, SNS, EventBridge
 Observability: Datadog (primary), AWS CloudWatch, OpenTelemetry, Jaeger traces
 Security: AWS KMS (BYOK), Secrets Manager, WAF, Shield Advanced, CrowdStrike Falcon
 IaC: Terraform 1.6+ (modules registry), Crossplane for K8s-native resources
 CI/CD: GitLab CI (self-hosted runners on EKS), ArgoCD, Flux for GitOps
 Compliance: AWS Config conformance packs, custom Lambda rules, Qualys VMDR

13. Cf. *supra*, Section 1.9, para. 25; see also Annex C ref. 1013/2026-EU.

CASE STUDY 2: AXA GROUP — "ONE CLOUD" KUBERNETES TRANSFORMATION

BEFORE: Pre-transformation (2022)	AFTER: "ONE CLOUD" Target (2026)
• 4 private clouds (AXA Technology Services legacy) • VMware-based virtualization (8,000 VMs) • Siloed per-entity infrastructure (AXA France, AXA XL, AXA IM) • 200+ distinct monitoring tools across entities • Average provisioning time: 3 weeks • Security: traditional SOC, SIEM (QRadar) • Compliance: manual processes, Excel-based tracking • 18 months average for new product launch • Annual IT spend: EUR 4.8B globally • Staffing: 12,000 IT FTEs, 60% outsourced	• GCP (primary) + Azure (M365/identity) + AWS (AXA XL) • 95% containerized on GKE/Anthos (target) • Unified platform: "AXA Cloud Platform" (ACP) — internal PaaS • Single observability: Dynatrace + Grafana/Loki/Tempo • Provisioning: <15 minutes (self-service portal) • Zero-trust: BeyondCorp, Zscaler, Palo Alto Prisma • Compliance: Policy-as-Code (OPA/Gatekeeper) • 6 weeks new product launch (target) • IT spend reduction: -25% by 2027 • In-house ratio: 70% internal by 2026

AXA SA (SIREN 572 093 920), 25 Avenue Matignon, 75008 Paris. Global workforce: ~145,000 employees (FY2024). Total revenue: EUR 102.7 billion (GWP). CTO: Guillaume BESSE (AXA Group Operations).

AXA's cloud journey represents one of the most ambitious Kubernetes adoption programs in European financial services. The "ONE CLOUD" initiative, launched in Q1 2023, aims to consolidate 40+ distinct infrastructure environments across 50 countries onto a unified cloud-native platform built primarily on Google Cloud Platform (GCP) with Google Kubernetes Engine (GKE) and Anthos for hybrid management.

"Kubernetes est devenu le système d'exploitation de l'entreprise. Avec plus de 2,500 clusters GKE en production et 180,000 pods actifs, nous opérons l'une des plus grandes installations Kubernetes du secteur financier mondial. Notre objectif est d'atteindre 95% de containerisation d'ici 2026, ne conservant que les mainframes legacy en bare-metal."
— Guillaume BESSE, CTO AXA Group Operations (KubeCon EU 2025 keynote, London)

Zero-Trust Security Architecture:
AXA has implemented a comprehensive zero-trust security model across its cloud estate, one of the first insurers globally to achieve this at scale. Key components:

- Identity & Access: Okta (workforce identity), Auth0 (customer identity), GCP IAM with Workload Identity Federation
- Network: Zscaler ZIA/ZPA (replacing all VPNs by Q2 2026), GCP VPC Service Controls, Private Service Connect
- Endpoint: CrowdStrike Falcon, Microsoft Defender for Endpoint, Jamf (Mac fleet)
- Application: Istio service mesh (mutual TLS), OPA/Gatekeeper admission control, Trivy vulnerability scanning
- Data: Google CMEK (Customer-Managed Encryption Keys), Thales CipherTrust for key management, DLP API
- Monitoring: Chronicle SIEM (Google), Cortex XSOAR (Palo Alto) for SOAR, custom detection rules (2,400+ active)

Compliance & Regulatory: ACPR/BCE supervision, DORA (Digital Operational Resilience Act) compliance program launched Q3 2024. AXA has allocated EUR 45M specifically for DORA readiness, including ICT risk management framework, third-party risk management for cloud providers, digital operational resilience testing (TLPT per ECB framework).

Hiring: AXA is actively recruiting across its technology division with 850+ open positions globally in cloud/platform engineering. France-specific: ~280 positions including:
— Kubernetes Platform Engineers (x55): GKE, Anthos, Istio, ArgoCD. Paris + Nantes.
— SRE / Observability Engineers (x30): Dynatrace, Grafana stack, SLO/SLI design.
— Cloud Security Architects (x25): Zero-trust design, GCP security, Zscaler.
— ML Engineers (x40): Vertex AI, TensorFlow, claims automation models.
— Data Platform Engineers (x35): BigQuery, Dataflow, Pub/Sub, dbt.

21. Cf. supra, Section 4.7, para. 31; see also Annex A ref. 1021/2025-EU
22. Cf. supra, Section 10.9, para. 13; see also Annex B ref. 1022/2026-EU
23. Cf. supra, Section 8.3, para. 38; see also Annex C ref. 1023/2025-EU
24. Cf. supra, Section 7.1, para. 22; see also Annex D ref. 1024/2025-EU

CASE STUDY 3: CREDIT AGRICOLE — HYBRID CLOUD & DORA COMPLIANCE

BEFORE: Pre-Cloud (2021)	AFTER: Hybrid Target (2027)
• 100% on-premise (SILCA data centres, Montrouge + Chartres) • Bull/Atos mainframes + x86 VMware estate • 22,000 applications across 39 Caisses Regionales • Average application age: 11.4 years • Deployment: quarterly release windows • Zero cloud presence (board-level cloud moratorium until 2021) • IT budget: EUR 5.1B/year (highest in French banking) • 28,000 IT staff (SILCA + CA-GIP + entities) • IBM MQ / TIBCO messaging (legacy ESB) • Oracle RAC databases: 3,400 instances	• Azure (primary cloud) + on-premise (sovereign workloads) • Azure Arc for hybrid management across all sites • 40% cloud by 2027 (conservative vs. peers) • OpenShift 4.x (Red Hat) on Azure + on-prem • Azure DevOps CI/CD (replacing Jenkins/SVN) • Azure Sentinel + Defender for Cloud (SIEM/XDR) • Deployment: weekly sprints, daily for cloud-native • "Cloud de Confiance" partnership (Bleu — Orange/Capgemini/MS) • Azure Cosmos DB, Azure SQL, PostgreSQL Flexible • Kafka (Confluent) replacing MQ/TIBCO

Credit Agricole SA (SIREN 784 608 416), 12 Place des Etats-Unis, 92127 Montrouge. Group structure: Federated model with 39 Caisses Regionales (cooperative banks), Amundi (asset management), Caceis (custody), LCL, CA-CIB (corporate/investment banking). Total group employees: ~147,000. Revenue (NBI): EUR 38.1 billion (FY2024).

Credit Agricole's cloud strategy is notably more conservative than Societe Generale's or AXA's, reflecting its cooperative governance structure and the complexity of coordinating across 39 semi-autonomous regional entities. The group's IT subsidiary, CA-GIP (Credit Agricole - Group Infrastructure Platform), manages centralized infrastructure from Montrouge and Chartres.

"Notre approche du cloud est pragmatique et souveraine. Nous ne migrons pas pour suivre une mode — chaque workload fait l'objet d'une analyse risque/bénéfice rigoureuse. Les données sensibles de nos 52 millions de clients resteront dans des infrastructures souveraines, qu'il s'agisse de nos datacentres ou du Cloud de Confiance." — Pierre-Antoine VACHERON, Directeur Général de CA-GIP (Les Echos, Oct. 2025)

DORA COMPLIANCE — REGULATION (EU) 2022/2554

The Digital Operational Resilience Act (DORA) entered into application on 17 January 2025, imposing stringent ICT risk management requirements on all EU financial entities. Credit Agricole's DORA compliance program is one of the most complex in French banking due to the federated structure:

- Chapter II — ICT Risk Management Framework:
- Appointed group-level CISO: Marc BRESSAND (since 2024)
 - ICT risk taxonomy: 847 identified risks across 6 categories
 - Business impact analysis: completed for all 22,000 applications (18-month effort)
 - Recovery Time Objective: <2h for critical functions, <24h for important functions
- Chapter V — ICT Third-Party Risk Management:
- Register of all ICT third-party providers: 2,340 vendors catalogued
 - Critical ICT service providers designated: Microsoft (Azure), Accenture (managed services), IBM (mainframe), Broadcom (VMware), Red Hat (OpenShift)
 - Concentration risk assessment: Azure dependency flagged as HIGH (mitigation: Bleu sovereign cloud, on-prem failover)
 - Annual due diligence reviews for all critical providers (Art. 28-30)
- Chapter IV — Digital Operational Resilience Testing:
- TLPT (Threat-Led Penetration Testing) per ECB/TIBER-EU framework
 - First TLPT cycle: Q2-Q4 2025 (scope: core banking, payments, trading)
 - Testing provider: Mandiant (Google) + internal Red Team (CA-GIP CSIRT)
 - Budget allocated: EUR 18M for 2025 testing cycle

Estimated total DORA compliance cost: EUR 95M over 2024-2026 (group-wide)

Hiring: CA-GIP is recruiting 200+ positions specifically for cloud and DORA-related roles:

- Azure Cloud Architects (x30): Landing zone, Azure Policy, Blueprints. Montrouge. 65-90k EUR.
- OpenShift/K8s Engineers (x25): OpenShift 4.14+, Operators, GitOps. Chartres + remote. 55-75k EUR.
- Cybersecurity Analysts (x40): Azure Sentinel, SOAR, threat hunting. Montrouge. 50-70k EUR.
- ICT Risk Officers (x20): DORA Art.5-16 implementation, EBA guidelines. 60-80k EUR.
- Data Engineers (x30): Azure Synapse, Data Factory, Databricks. 55-80k EUR.
- DevOps Engineers (x35): Azure DevOps, Terraform, Ansible, Packer. 50-70k EUR.
- Remaining: project managers, compliance officers, change management.

31. Cf. supra, Section 9.4, para. 8; see also Annex A ref. 1031/2024-EU

32. Cf. supra, Section 11.8, para. 17; see also Annex B ref. 1032/2026-EU

33. Cf. supra, Section 5.1, para. 26; see also Annex C ref. 1033/2026-EU

34. Cf. supra, Section 12.6, para. 9; see also Annex D ref. 1034/2026-EU

35. Cf. supra, Section 1.8, para. 1; see also Annex E ref. 1035/2024-EU

CONSOLIDATED COMPARISON MATRIX

Dimension	Societe Generale	AXA	Credit Agricole
Primary Cloud	AWS	GCP	Azure
Secondary Cloud	Azure	Azure + AWS	On-premise (sovereign)
Container Platform	Amazon EKS	GKE / Anthos	OpenShift 4.x
Employees (Global)	~117,000	~145,000	~147,000
IT Budget (est.)	EUR 3.2B	EUR 4.8B	EUR 5.1B
Cloud Migration %	65% by 2026	95% by 2026	40% by 2027
Cloud Hiring (FR)	~400 positions	~280 positions	~200 positions
DORA Budget	EUR 55M	EUR 45M	EUR 95M
IaC Tool	Terraform	Terraform + Crossplane	Terraform + ARM/Bicep
CI/CD	GitLab CI	GitLab CI + ArgoCD	Azure DevOps
Observability	Datadog	Dynatrace	Azure Monitor + Elastic
Security Model	Zero-trust (partial)	Zero-trust (BeyondCorp)	Perimeter + zero-trust (hybrid)
SIEM	Splunk + AWS SecurityHub	Chronicle (Google)	Azure Sentinel
Key Challenge	Oracle DB migration	Multi-entity consolidation	Federated governance
Sovereign Cloud	S3NS (Thales/Google)	N/A (GCP direct)	Bleu (Orange/Capgemini/MS)

41. Cf. supra, Section 1.9, para. 22; see also Annex A ref. 1041/2026-EU
42. Cf. supra, Section 7.8, para. 3; see also Annex B ref. 1042/2024-EU
43. Cf. supra, Section 2.2, para. 20; see also Annex C ref. 1043/2025-EU